

Lithium Iron Phosphate (LiFePO4) batteries are increasingly popular due to their high energy density, long cycle life, and safety features.

This guide provides an overview of **LiFePO4 battery voltage**, the concept of **battery state of charge(SOC)**, and voltage charts corresponding to common LiFePO4 battery specifications, along with reference **tables for battery voltage and SOC**.

Due to their charging method differing from other types of batteries, please refer to our **separate article** for details on LiFePO4 battery charging voltage parameters, which covers **how to charge LiFePO4 batteries and configure charging parameters** comprehensively.

Quick Navigation

1. What is the Nominal Voltage LiFePO4 Battery
2. What is Battery Stage of Charge - SOC
3. LiFePO4 Battery Voltage Chart
 - 12V LiFePO4 Battery Voltage Chart
 - 24V LiFePO4 Battery Voltage Chart
 - 48V LiFePO4 Battery Voltage Chart
4. How to Check LiFePO4 SoC and Battery Voltage

What is the Nominal Voltage LiFePO4 Battery

Nominal voltage is commonly used to describe the battery's characteristics, tested under **standard conditions: 25°C temperature, 50% charge, and moderate load**, although the actual voltage can fluctuate depending on the charge level.

A **LiFePO4 battery cell** typically has a **nominal voltage** of 3.2 volts, helps in comparing and designing systems. However, a **fully charged LiFePO4 cell** might have a voltage of around **3.6 to 3.65 volts**, while a fully discharged cell might drop to around 2.5 to 2.8 volts. These cells are the **fundamental building blocks** of any LiFePO4 battery pack.

For higher voltage or capacity, these cells are connected **in series (denoted as "s")**, their voltages add up, forming the total voltage of the battery pack.

For example, a **12V LiFePO4 battery** is composed of 4 cells in series (4s), resulting in a nominal voltage of 12.8V ($4 \times 3.2V$). When fully charged, each cell reaches around 3.65V, making the fully charged voltage of a 12V battery approximately 14.6V. Similarly, a **24V battery pack** usually consists of 8 cells in series (8s), yielding a nominal voltage of 25.6V, and when fully charged, the voltage reaches around 29.2V. A **48V battery pack** generally uses 16 cells in series (16s), giving a nominal voltage of 51.2V, and when fully charged, it reaches around 58.4V.

It's important to note the difference between a **15s and 16s configuration for a 48V system**. A 15s pack, while sometimes used, has a slightly lower nominal voltage of 48V ($3.2V \times 15$), with a fully charged voltage of around 54.75V, which can cause compatibility issues with some devices. The more common 16s configuration offers a nominal voltage of 51.2V and a fully charged voltage of around 58.4V, aligning better with standard 48V equipment.

By combining the battery pack in parallel or series connection in battery group, user can make a battery backup system for even greater capacity or desired energy density.

What is Battery Stage of Charge - SOC

Contrary to battery Depth of Discharge (DoD), Stage of Charge (SoC) shows the **remaining charge level** as a percentage of the battery's total capacity. In essence, "SoC = 100% - DoD". For example, if a battery has a recommended DoD of 80% for a lifespan of 6000 cycles, it means you should charge the battery when its SoC drops to 20% or lower.

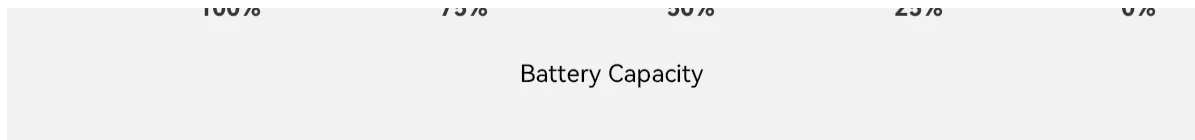
LiFePO4 Battery Voltage Chart

A voltage chart is commonly used to monitor the State of Charge (SOC) of a LiFePO4 battery.

Going with the LiFePO4 battery charging stages, voltage varies from a **rapid increase** during the bulk stage to a **slower rise** during the absorption stage. As the battery approaches full charge, the voltage **plateaus** around 3.6 to 3.7 volts per cell, as illustrated in the battery voltage curve below.

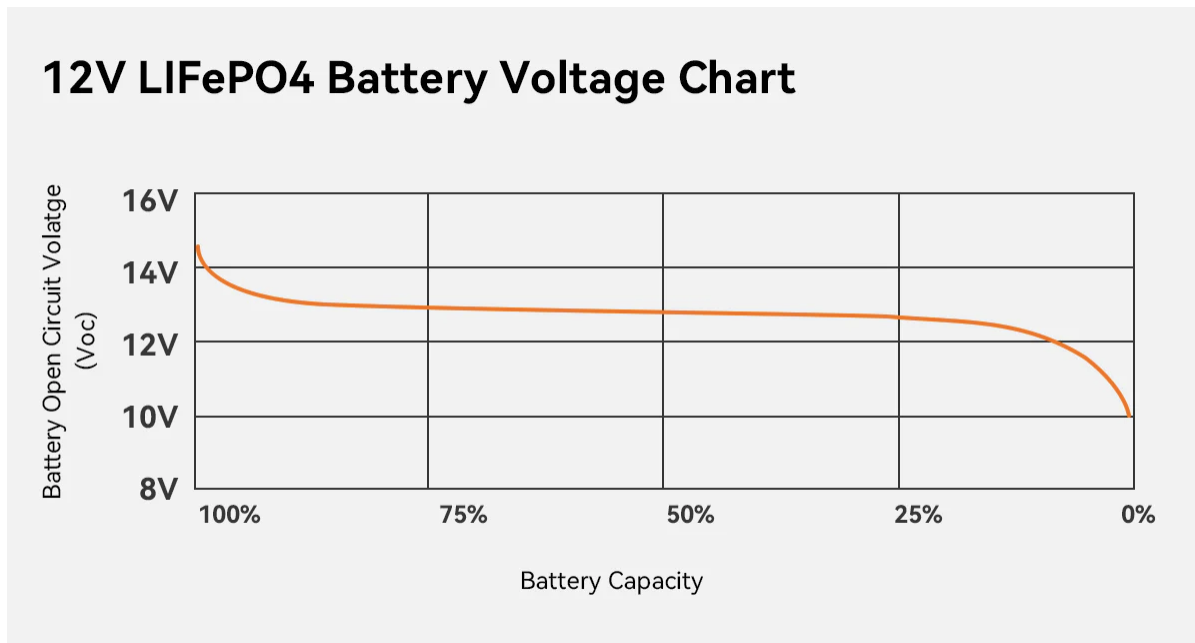
3.2 LiFePO4 Cell Voltage Chart





By comparing the battery's voltage to the standardized chart, users can **estimate the remaining capacity**. Below, we provide voltage charts for 12V, 24V, and 48V LiFePO4 batteries.

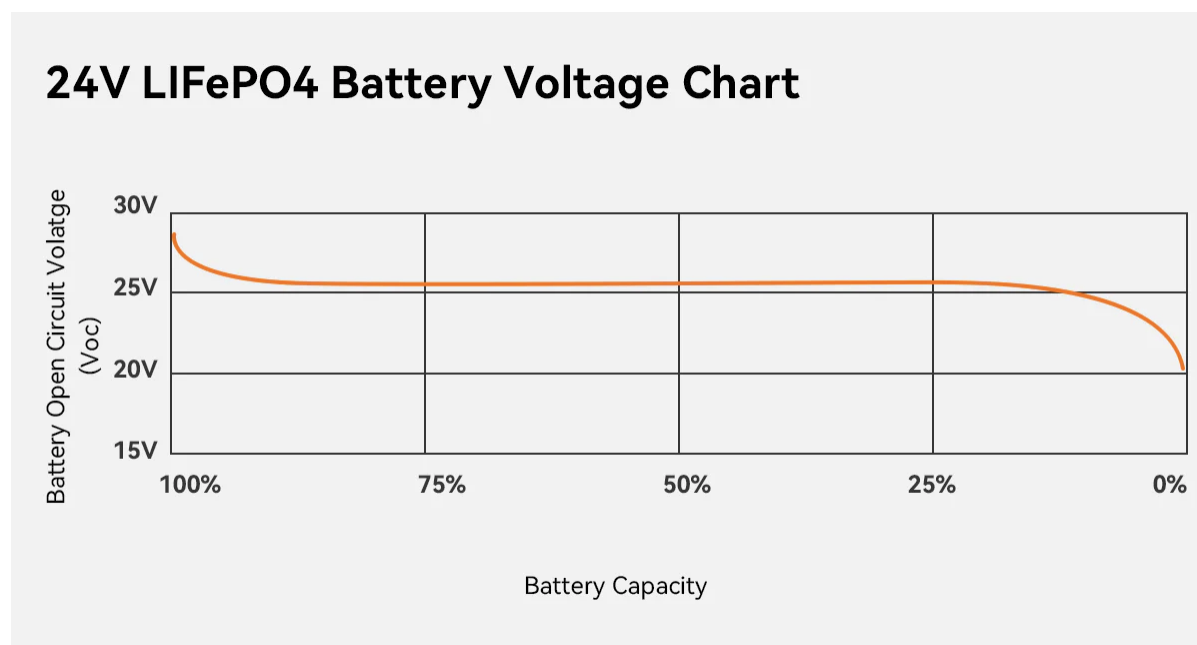
12V LiFePO4 Battery Voltage Chart



SoC%	Volt per Cell	12V (4s)
100.00%	3.65	14.6
99.50%	3.45	13.8
99.00%	3.38	13.52
90.00%	3.35	13.4
80.00%	3.33	13.32
70.00%	3.30	13.2
60.00%	3.28	13.12
50.00%	3.26	13.04

40.00%	3.25	13.00
30.00%	3.23	12.92
20.00%	3.20	12.8
15.00%	3.05	12.2
9.50%	3.00	12.0
5.00%	2.80	11.2
0.50%	2.54	10.16
0.00%	2.50	10.0

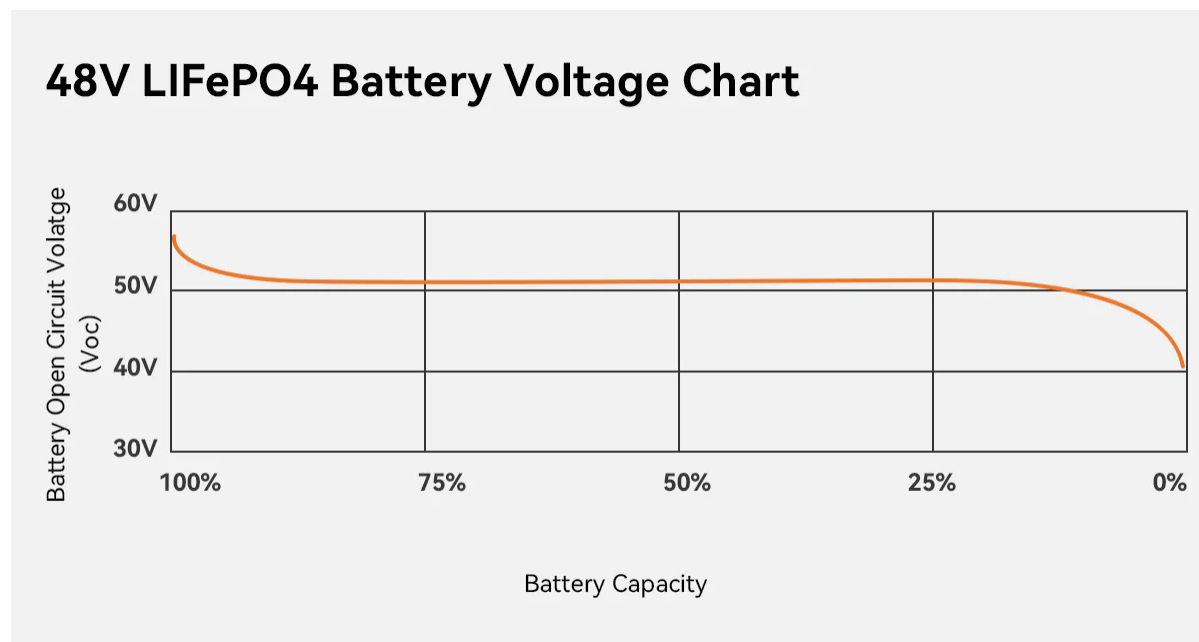
24V LiFePO4 Battery Voltage Chart



SoC%	Volt per Cell	24V (8s)
100.00%	3.65	29.2
99.50%	3.45	27.6
99.00%	3.38	27.04
90.00%	3.35	26.8
80.00%	3.33	26.64

70.00%	3.30	26.4
60.00%	3.28	26.24
50.00%	3.26	26.08
40.00%	3.25	26
30.00%	3.23	25.84
20.00%	3.20	25.6
15.00%	3.05	24.4
9.50%	3.00	24
5.00%	2.80	22.4
0.50%	2.54	20.32
0.00%	2.50	20

48V LiFePO4 Battery Voltage Chart



SoC%	Volt per Cell	48V (15s)	51.2V (16s)
100.00%	3.65	54.75	58.4
99.50%	3.45	51.75	55.2

99.00%	3.38	50.7	54.08
90.00%	3.35	50.25	53.6
80.00%	3.33	49.95	53.28
70.00%	3.3	49.5	52.8
60.00%	3.28	49.2	52.48
50.00%	3.26	48.9	52.16
40.00%	3.25	48.75	52
30.00%	3.23	48.45	51.68
20.00%	3.2	48	51.2
15.00%	3.05	45.75	48.8
9.50%	3	45	48
5.00%	2.8	42	44.8
0.50%	2.54	38.1	40.64
0.00%	2.5	37.5	40

How to Check LiFePO4 SoC and Battery Voltage

Using battery with monitor

Some LiFePO4 batteries, like PowMr 51.2v battery, often come equipped with **built-in monitor and indicator** that display both voltage and State of Charge (SoC). This feature allows you to quickly and easily check the battery's status without additional tools. Simply view the information on the display, which provides real-time data about your battery's performance.

Using a Multimeter

To measure the open circuit voltage of a LiFePO4 battery using a multimeter, follow these steps for a moderately accurate result. Keep in mind that you'll need to **disconnect all loads and chargers** and let the battery rest.